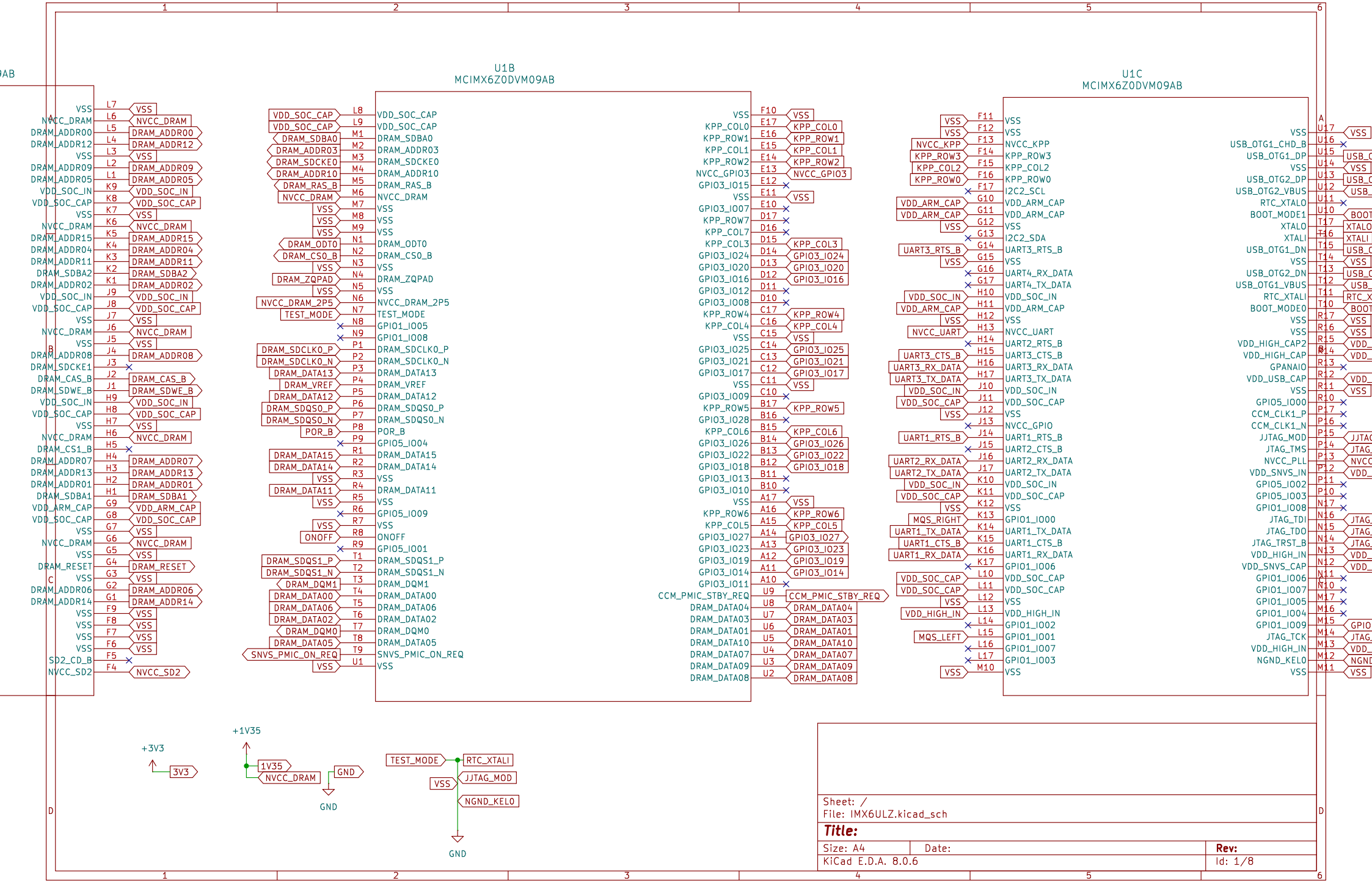
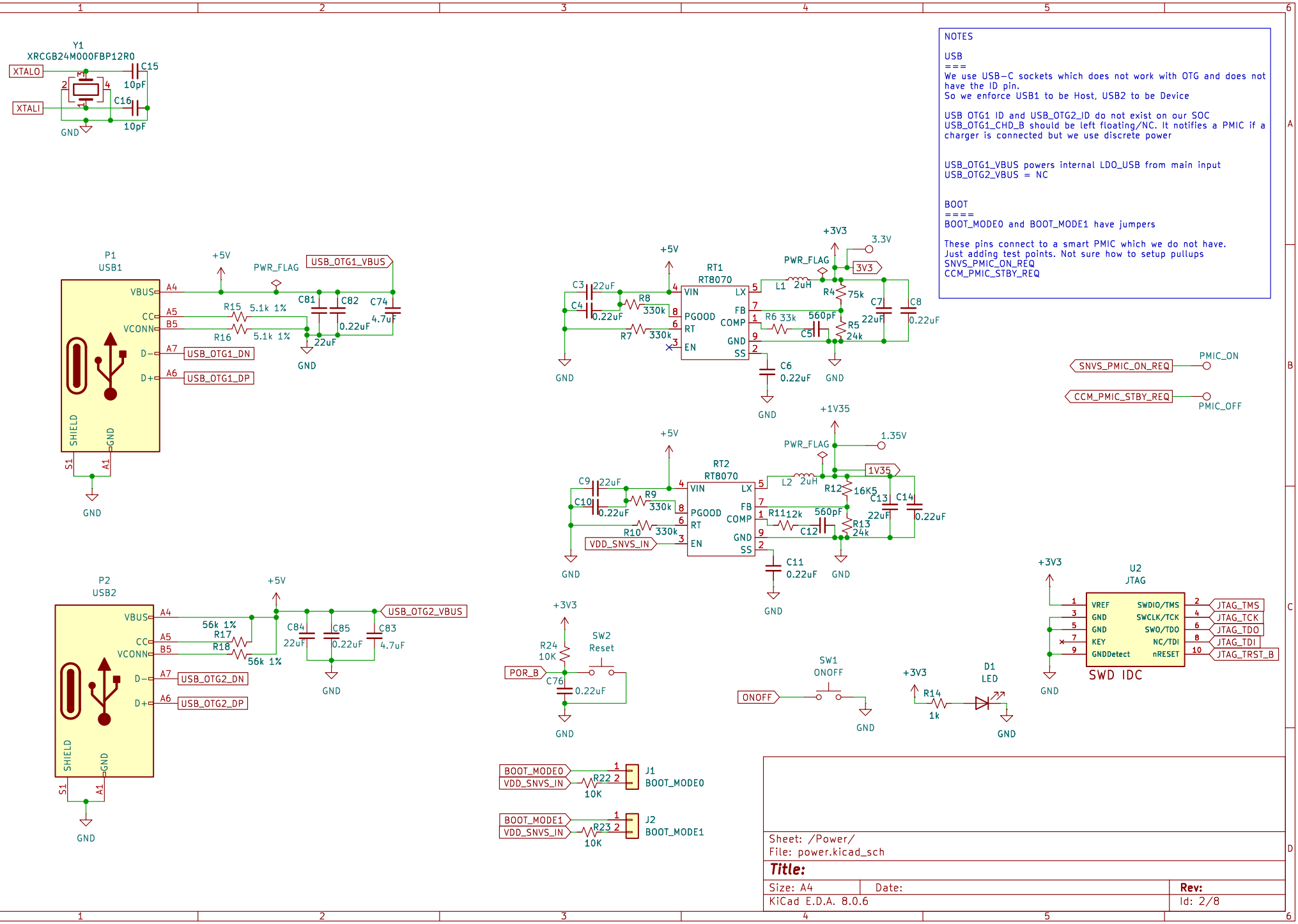
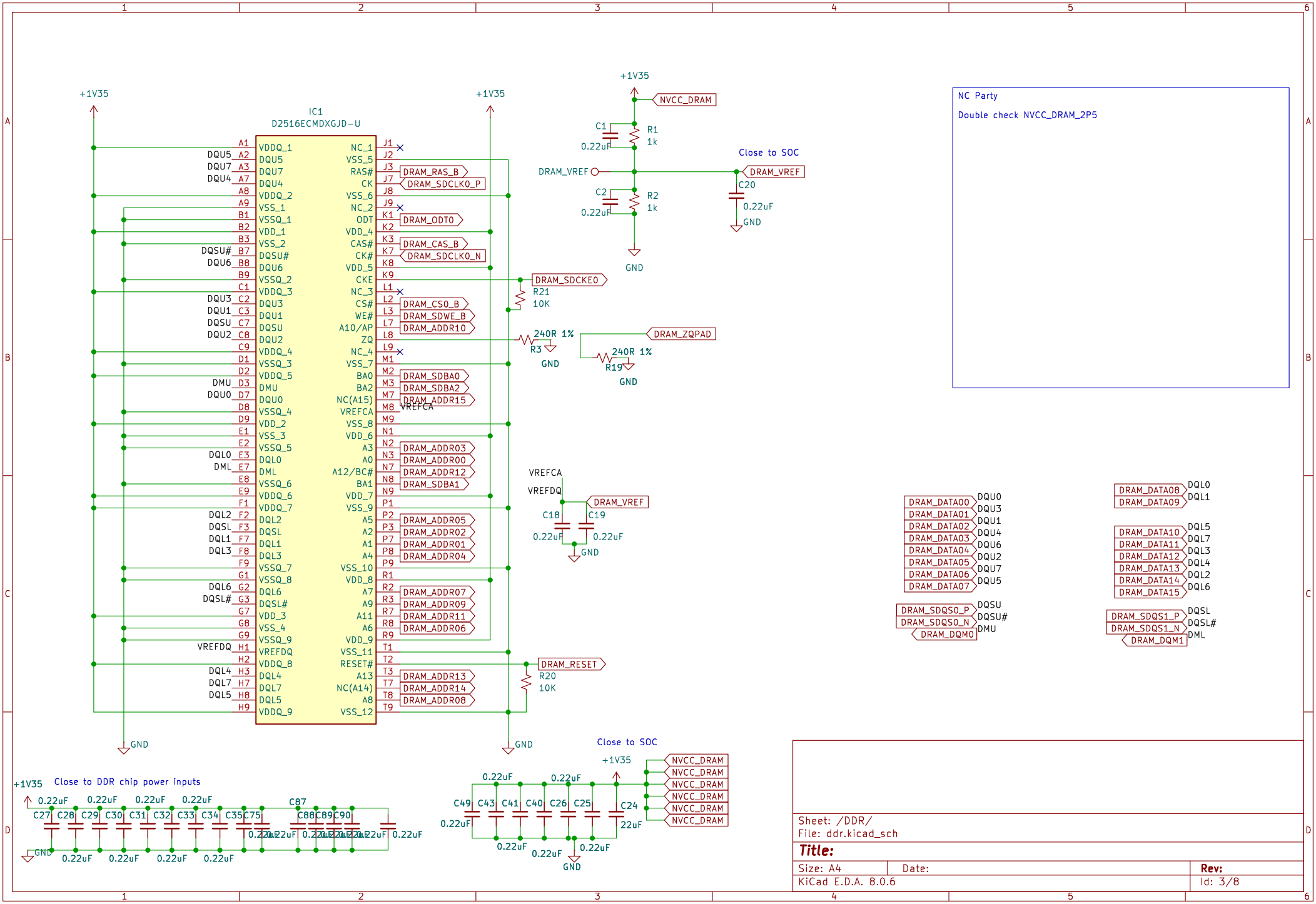
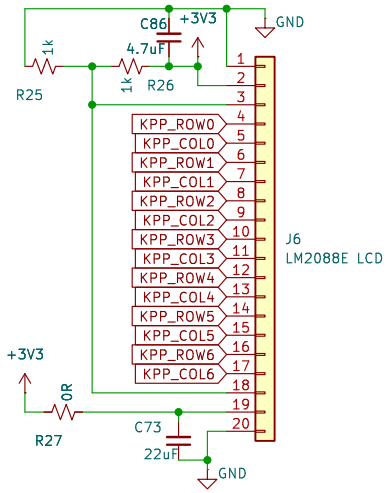
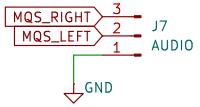

U1B
MCIMX6Z0DVM09AB
U1C
MCIMX6Z0DVM09AB
VSS
NVCC_DRAM
DRAM_ADDR00
DRAM_ADDR12
DRAM_ADDR09
DRAM_ADDR05
VDD_SOC_IN
VDD_SOC_CAP
DRAM_ADDR15
DRAM_ADDR04
DRAM_ADDR11
DRAM_SDBA2
DRAM_ADDR02
DRAM_ADDR08
DRAM_SDCKE1
DRAM_CAS_B
DRAM_SDWE_B
DRAM_CS1_B
DRAM_ADDR07
DRAM_ADDR13
DRAM_ADDR01
DRAM_SDBA1
VDD_ARM_CAP
DRAM_RESET
DRAM_ADDR06
DRAM_ADDR14
SD2_CD_B
NVCC_SD2
VDD_SOC_CAP
DRAM_SDBA0
DRAM_ADDR03
DRAM_SDCKE0
DRAM_ADDR10
DRAM_RAS_B
DRAM_ODT0
DRAM_CS0_B
DRAM_ZQPAD
NVCC_DRAM_2P5
TEST_MODE
GPIO1_IO05
GPIO1_IO08
DRAM_SDCLK0_P
DRAM_SDCLK0_N
DRAM_DATA13
DRAM_VREF
DRAM_DATA12
DRAM_SDQS0_P
DRAM_SDQS0_N
POR_B
GPIO5_IO04
DRAM_DATA15
DRAM_DATA14
DRAM_DATA11
GPIO5_IO09
ONOFF
GPIO5_IO01
DRAM_SDQS1_P
DRAM_SDQS1_N
DRAM_DQM1
DRAM_DATA00
DRAM_DATA06
DRAM_DATA02
DRAM_DQM0
DRAM_DATA05
SNVS_PMIC_ON_REQ
KPP_COL0
KPP_ROW1
KPP_COL1
KPP_ROW2
NVCC_GPIO3
GPIO3_IO15
GPIO3_IO07
KPP_ROW7
KPP_COL7
KPP_COL3
GPIO3_IO24
GPIO3_IO20
GPIO3_IO16
GPIO3_IO12
GPIO3_IO08
KPP_ROW4
KPP_COL4
GPIO3_IO25
GPIO3_IO21
GPIO3_IO17
GPIO3_IO09
KPP_ROW5
GPIO3_IO28
KPP_COL6
GPIO3_IO26
GPIO3_IO22
GPIO3_IO18
GPIO3_IO13
GPIO3_IO10
KPP_ROW6
KPP_COL5
GPIO3_IO27
GPIO3_IO23
GPIO3_IO19
GPIO3_IO14
GPIO3_IO11
CCM_PMIC_STBY_REQ
DRAM_DATA04
DRAM_DATA03
DRAM_DATA01
DRAM_DATA10
DRAM_DATA07
DRAM_DATA09
DRAM_DATA08
NVCC_KPP
KPP_ROW3
KPP_COL2
KPP_ROW0
I2C2_SCL
VDD_ARM_CAP
I2C2_SDA
UART3_RTS_B
UART4_RX_DATA
UART4_TX_DATA
VDD_SOC_IN
NVCC_UART
UART2_RTS_B
UART3_CTS_B
UART3_RX_DATA
UART3_TX_DATA
NVCC_GPIO
UART1_RTS_B
UART2_CTS_B
UART2_RX_DATA
UART2_TX_DATA
MQS_RIGHT
UART1_TX_DATA
UART1_CTS_B
UART1_RX_DATA
GPIO1_IO06
MQS_LEFT
GPIO1_IO02
GPIO1_IO01
GPIO1_IO07
GPIO1_IO03
USB_OTG1_CHD_B
USB_OTG1_DP
USB_OTG2_DP
USB_OTG2_VBUS
RTC_XTALO
BOOT_MODE1
XTALO
XTALI
USB_OTG1_DN
USB_OTG2_DN
USB_OTG1_VBUS
RTC_XTALI
BOOT_MODE0
VDD_HIGH_CAP2
VDD_HIGH_CAP
GPANAIO
VDD_USB_CAP
GPIO5_IO00
CCM_CLK1_P
CCM_CLK1_N
JTAG_MOD
JTAG_TMS
NVCC_PLL
VDD_SNVS_IN
GPIO5_IO02
GPIO5_IO03
GPIO1_IO08
JTAG_TDI
JTAG_TDO
JTAG_TRST_B
VDD_HIGH_IN
VDD_SNVS_CAP
GPIO1_IO06
GPIO1_IO07
GPIO1_IO05
GPIO1_IO04
GPIO1_IO09
JTAG_TCK
VDD_HIGH_IN
NGND_KEL0
+3V3
3V3
+1V35
1V35
NVCC_DRAM
GND
TEST_MODE
RTC_XTALI
VSS
JJTAG_MOD
NGND_KEL0
Sheet: /
File: IMX6ULZ.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.6
Id: 1/8

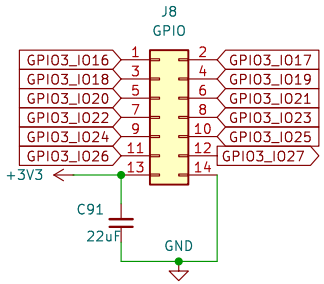

NOTES
USB
===
We use USB-C sockets which does not work with OTG and does not have the ID pin.
So we enforce USB1 to be Host, USB2 to be Device
USB OTG1 ID and USB_OTG2_ID do not exist on our SOC
USB_OTG1_CHD_B should be left floating/NC. It notifies a PMIC if a charger is connected but we use discrete power
USB_OTG1_VBUS powers internal LDO_USB from main input
USB_OTG2_VBUS = NC
BOOT
====
BOOT_MODE0 and BOOT_MODE1 have jumpers
These pins connect to a smart PMIC which we do not have.
Just adding test points. Not sure how to setup pullups
SNVS_PMIC_ON_REQ
CCM_PMIC_STBY_REQ
Y1
XRCGB24M000FBP12R0
XTALO
XTALI
C15
10pF
C16
10pF
GND
P1
USB1
VBUS
CC
VCONN
D-
D+
SHIELD
GND
A4
A5
B5
A7
A6
S1
A1
+5V
PWR_FLAG
USB_OTG1_VBUS
R15 5.1k 1%
R16 5.1k 1%
C81
C82
0.22uF
C74
4.7uF
22uF
USB_OTG1_DN
USB_OTG1_DP
P2
USB2
USB_OTG2_VBUS
R17 56k 1%
R18 56k 1%
C84 22uF
C85 0.22uF
C83 4.7uF
USB_OTG2_DN
USB_OTG2_DP
RT1
RT8070
VIN
LX
FB
PGOOD
COMP
RT
GND
EN
SS
C3 22uF
C4 0.22uF
R8 330k
R7 330k
L1 2uH
R4 75k
R6 33k
C5 560pF
R5 24k
C7 22uF
C8 0.22uF
C6 0.22uF
3.3V
3V3
+3V3
RT2
RT8070
C9 22uF
C10 0.22uF
R9 330k
R10 330k
VDD_SNVS_IN
L2 2uH
R12 16K5
R11 12k
C12 560pF
R13 24k
C13 22uF
C14 0.22uF
C11 0.22uF
1.35V
1V35
+1V35
SNVS_PMIC_ON_REQ
CCM_PMIC_STBY_REQ
PMIC_ON
PMIC_OFF
R24 10K
POR_B
C76 0.22uF
SW2
Reset
SW1
ONOFF
R14 1k
D1
LED
U2
JTAG
VREF
SWDIO/TMS
GND
SWCLK/TCK
GND
SWO/TDO
KEY
NC/TDI
GNDDetect
nRESET
JTAG_TMS
JTAG_TCK
JTAG_TDO
JTAG_TDI
JTAG_TRST_B
SWD IDC
BOOT_MODE0
VDD_SNVS_IN
R22 10K
J1
BOOT_MODE0
BOOT_MODE1
R23 10K
J2
BOOT_MODE1
Sheet: /Power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.6
Id: 2/8

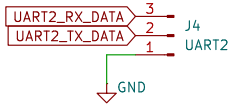

IC1
D2516ECMDXGJD-U
+1V35
GND
NVCC_DRAM
Close to SOC
DRAM_VREF
C1
0.22uF
R1
1k
C2
0.22uF
R2
1k
C20
0.22uF
DRAM_SDCKE0
R21
10K
240R 1%
R3
DRAM_ZQPAD
240R 1%
R19
VREFCA
VREFDQ
C18
C19
0.22uF
0.22uF
DRAM_RESET
R20
10K
NC Party
Double check NVCC_DRAM_2P5
DRAM_DATA00
DRAM_DATA01
DRAM_DATA02
DRAM_DATA03
DRAM_DATA04
DRAM_DATA05
DRAM_DATA06
DRAM_DATA07
DRAM_DATA08
DRAM_DATA09
DRAM_DATA10
DRAM_DATA11
DRAM_DATA12
DRAM_DATA13
DRAM_DATA14
DRAM_DATA15
DRAM_SDQS0_P
DRAM_SDQS0_N
DRAM_DQM0
DRAM_SDQS1_P
DRAM_SDQS1_N
DRAM_DQM1
Close to DDR chip power inputs
C27
C28
C29
C30
C31
C32
C33
C34
C35
C75
C87
C88
C89
C90
Close to SOC
C49
C43
C41
C40
C26
C25
C24
22uF
Sheet: /DDR/
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 8.0.6
Id: 3/8

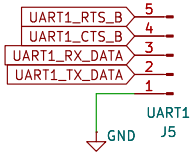
LM2088E pinout
Pin Name Descriptions
1 VSS 0V Power Supply, GND
2 VDD Positive Power Supply
3 V0 LCD Contrast Reference Input
4 /WR Write enable input, active LOW
5 /RD Read enable input, active LOW
6 /CS Chip Select Signal
/CS=LOW: Data IO is enabled
7 A0 Data Type Select
A0=H: command write, display data or cursor add read
A0=L: status flag read, display data or parameter write
8 /RES Reset Signal:
/RES = L, Reset the LCD Module
/RES = H, Normal Running
9 DB0
: :
16 DB7
8-bit bi-directional data bus
17 /WAIT Wait signal (*1)
18 VOUT Power Booster Output for V0
19 BLA Positive Power Supply for LED backlight
20 BLK Negative Power Supply for LED backlight











Sheet: /GPIO/
File: gpio.kicad_sch

Title:

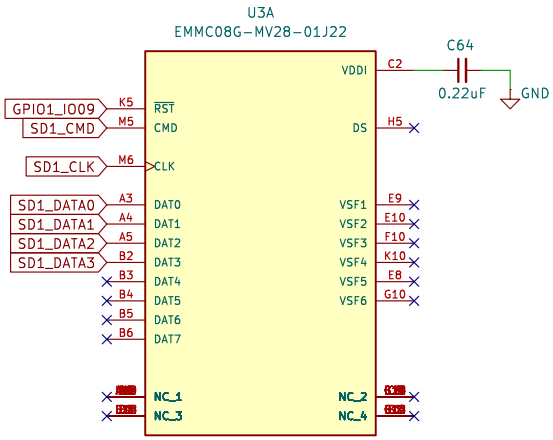
Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.6		Id: 4/8

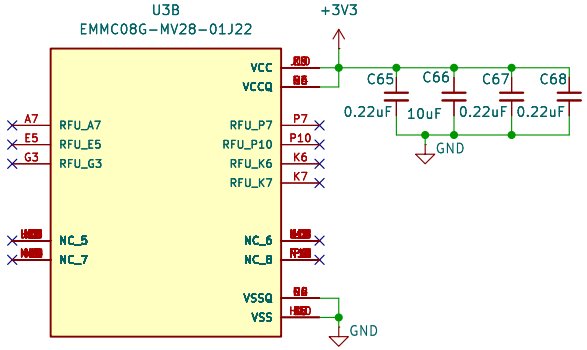
From Datasheet:
"Hardware Reset: By default, hardware reset is disabled and must be enabled in the EXT_CSD register if used. Otherwise, it can be left un-connected."

Data Stribe pin set to NC. It is used for HS400 mode

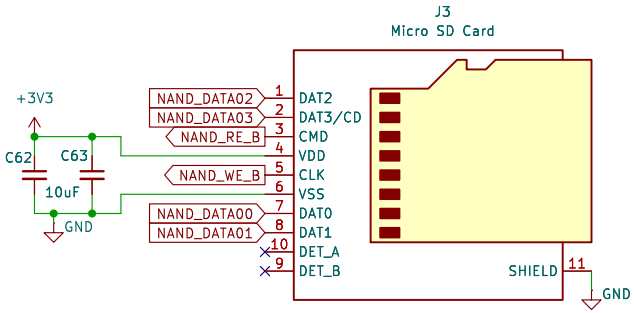
We'll just do 4pins because thats the max supported with 3.3V IO
With DDR and 52MHz that's 108MB/s

Ref:
https://www.mouser.ca/datasheet/2/1389/Kingston_05_02_2024_EMMC08G_WV28_01J22_v1_1-3451902.pdf





Sheet: /eMMC/		
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Size: A4	Date:	**Rev:**
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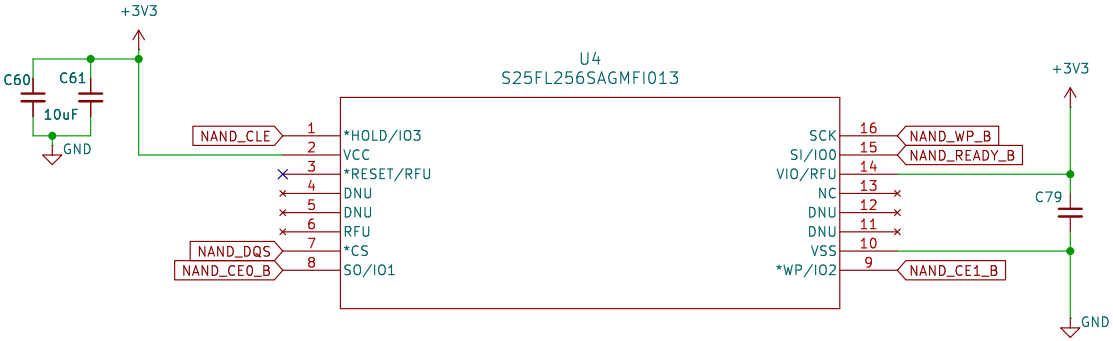


Should WP be grounded? It has 4 points on the footprint

Sheet: /SD/
File: sdcard.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.6		Id: 7/8



BALL	SIGNAL	MUX	
NAND_WP_B	qspi.A_SCLK	Alt2	
NAND_DQS	qspi.A_SS0_B	Alt2	
NAND_READY_B	qspi.A_DATA[0]	Alt2	
NAND_CE0_B	qspi.A_DATA[1]	Alt2	
NAND_CE1_B	qspi.A_DATA[2]	Alt2	
NAND_CLE	qspi.A_DATA[3]	Alt2	
NAND_DATA07	qspi.A_SS1_B	Alt2	(CS if CS is needed)

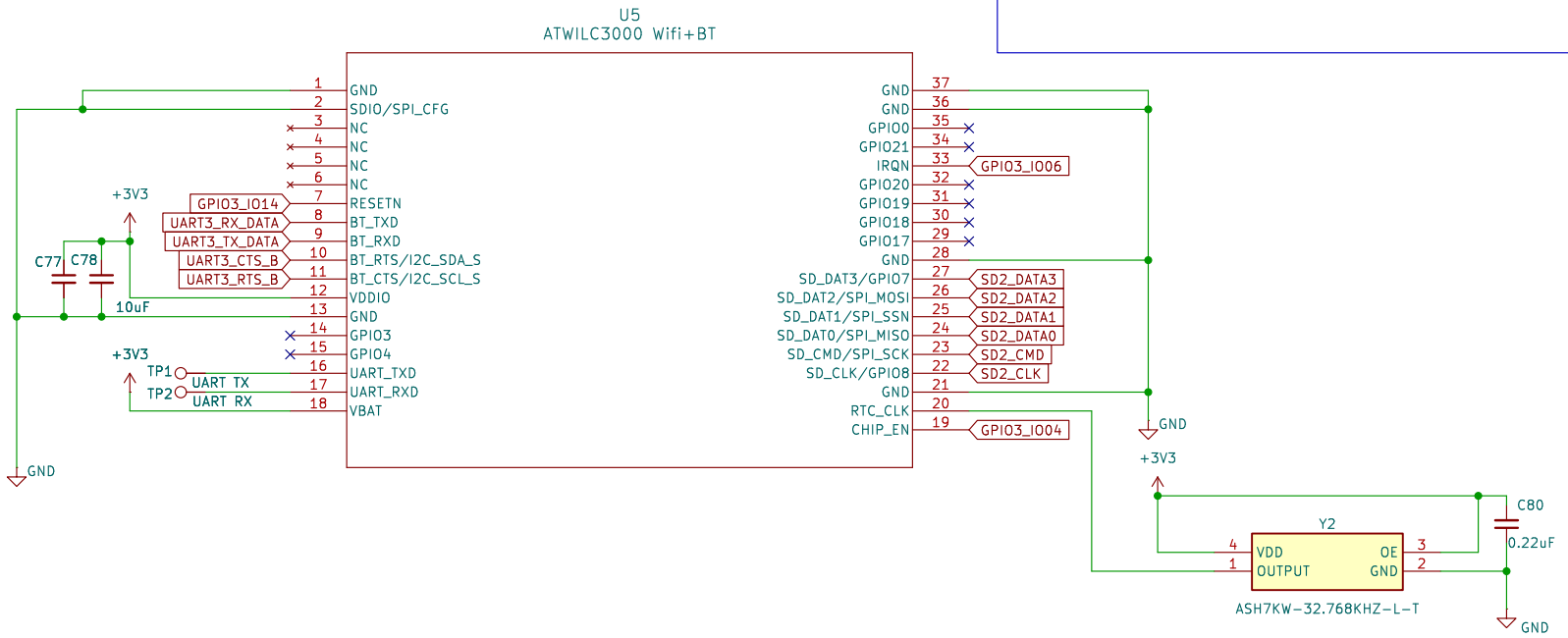
Sheet: /QSPI/
File: qspi.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.6		Id: 8/8

GPIO PINS used

INT (under sdmmc): GPIO3_IO06
RESETN: (under wifi_pwrseq): GPIO3_IO14
CHIP EN: (under wifi_pwrseq / powerdown-gpios): GPIO3_IO04



Sheet: /ATWILC3000/
File: atwilc3000.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 8.0.6		Id: 9/8